

Congestion Mitigation for Foraging Robot Swarms Using Adaptive Spiral Path Strategies

Arturo Gonzalez¹, Artemisa Trevino¹, and Qi Lu²

Abstract—Swarm robotics offers scalable and robust solutions for tasks such as foraging, yet congestion near central collection zones remains a critical challenge, especially with increasing swarm sizes. Traditional solutions, such as static path planning or local repulsion-based methods, often fail to prevent inter-robot collisions or bottlenecks near the collection zones. This research presents a comparative study of three strategies to mitigate congestion when returning resources to the central collection zone. The research herein focuses on tightly packed environments where, in theory, robots should follow a pre-planned spiral—either square or circular—with congestion detection as described in the first two strategies. The third strategy introduces an adaptive path that allows robots to make reactive movements in response to congestion. Furthermore, we explore the use of a deep reinforcement learning (DRL) approach that trains policies in a centralized manner but executes them in a decentralized fashion, thereby preserving swarm robotic principles. Both spiral strategies integrate multiple entry points into the central collection zone and dynamic re-routing upon congestion detection. Experimental evaluation in the ARGoS physics-based simulation environment demonstrates significant improvements in task completion time, collision reduction, and system throughput. These results indicate that structured congestion-aware trajectories can significantly improve swarm foraging efficiency.

I. INTRODUCTION

Swarm robotics focuses on coordinating multiple simple, independent robots to achieve complex tasks efficiently. They draw inspiration from natural systems like ants, termite networks, and flocks of birds. Current research in swarm robotics spans behaviors such as self-organization, aggregation, object sorting, and foraging. One of the most prominent areas of swarm robotics is foraging, where automated robots locate, gather, and transport resources to a designated collection zone. This animal behavior is inspired by ants hunting and delivering resources to local anthills, or formicary, to provide nutritional resources for the colony.

The collective nature of swarms allows for exploration in robustness, scalability, and parallel task execution. However, as the swarm size increases, congestion and inter-robot collisions become major bottlenecks, particularly near the collection zone. These issues can significantly impact overall resource collection efficiency, leading to delays, increased energy consumption, and potential damage to robots.

In swarm robotics, traditional foraging algorithms - such as the Centrally Placed Foraging Algorithm (CPFA) - often

operate under the assumption that increasing the number of robots in a swarm leads to greater efficiency [2]. However, real-world and simulation observations suggest otherwise - robots competing for limited space in the collection zone often experience congestion, forcing them to take inefficient detours or wait to access the drop-off area in the central collection zone. This phenomenon reduces the benefits of large-scale swarms and introduces challenges in maximizing efficiency.

An illustration of this phenomenon can be observed in ARGoS simulations [1]. 32 robots forage within an 8×8 arena (see Fig. 1). When many robots deliver collected resources to the central collection zone, time spent in collision avoidance increases dramatically.

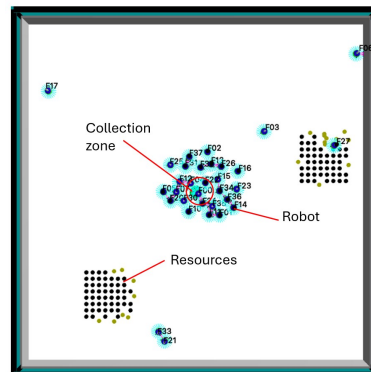


Fig. 1. An illustration of congestion around the central collection zone in ARGoS simulation. 32 robots are deployed in the 8×8 m arena. 128 resources are distributed in two 8×8 clusters. Some of them are collected and delivered to the center.

To address this problem, we propose an adaptive, decentralized congestion mitigation strategy inspired by biologically grounded movement analysis. By analyzing a robot's trajectory in real time, it can detect signs of congestion, such as erratic motion or reduced progress, and autonomously reroute to reduce collisions near the central collection zone. We evaluate this system using trajectory-based congestion detection metrics and implement structured and adaptive spiral movement patterns as recovery behaviors when congestion is detected. This approach allows robots to avoid compounding traffic near the drop-off zone and dynamically choose alternative entry points into the collection area.

In swarm robotics, tasks that may seem logical in simulation become fraught with practical challenges like terrain noise, limited or no communication, or high-density swarms. Our work demonstrates this theme as we observe a dramatic

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system degradation as robot density increases. These findings emphasize the importance of developing congestion-aware, robustly adaptive behavior that can scale to the dynamic, adversarial conditions faced in natural environments—whether in precision agriculture, forest monitoring, or outdoor search and rescue missions. By solely using local positional data, our method provides a scalable alternative to centralized planning or dense communication, both of which may be brittle in outdoor, bandwidth-limited domains.

II. RELATED WORKS

Congestion in swarm robotics, especially during central-place foraging tasks, has been increasingly studied in recent years. Many algorithms focus on improving task efficiency by reducing robot collisions and traffic bottlenecks. Spiral-based foraging algorithms have shown potential for efficient exploration. Aggarwal et al. [6] compared deterministic spiral methods to stochastic ant-inspired strategies, finding that spiral paths generally result in higher resource collection rates during simulations. However, their performance tends to decline when faced with real-world constraints like physical obstacles and sensor inaccuracies. Bharaswadkar et al. [7] further developed this concept with their Clustered-CPFA method, which introduces waypoint planning and delays to reduce robot interference in densely populated areas. Although this strategy effectively decreases congestion in clustered resources, it remains mostly reactive and relies heavily on prior knowledge of congested regions.

Other studies have addressed congestion through predictive and distributed control approaches. Marcolino et al. explored various methods, such as random delays, region separation, and hybrid models to manage robot competition near shared resources [4], [5]. While these methods smooth out robot arrival patterns, they can compromise overall efficiency by introducing deliberate delays or restricting movement areas. More recently, Passos et al. proposed algorithms like Single Queue Former (SQF) and Touch and Run Vector Fields (TRVF), which organize robots into queues or direct them using vector fields to maximize efficiency in high-demand areas [3]. Yet, these algorithms often depend on potential fields, making them vulnerable to issues like local minima, and their effectiveness can vary significantly based on robot density and area size. Hierarchical control strategies also play a role in congestion management. Vinicius and Chaimowicz developed an abstraction-based approach, creating elliptical structures to coordinate robot groups effectively, thus preventing navigation deadlocks in multi-group scenarios [8]. Despite being structured and reliable, these hierarchical methods can lack the adaptability needed in decentralized or quickly evolving environments.

Learning-based methods, specifically decentralized deep reinforcement learning (DRL), offer adaptive collision avoidance and dynamic path optimization. Long et al. [10] demonstrated that DRL policies at the sensor level could effectively handle unseen obstacles and varying robot densities, outperforming conventional rule-based methods in complex multi-robot scenarios. Nonetheless, DRL solutions require

extensive training data and computational resources, and their internal decision-making processes tend to be opaque, posing challenges for their use in safety-critical systems. Recent research also underscores the interconnectedness of congestion, scalability, and collective decision-making. Soma et al. [11] observed that larger swarm sizes can negatively impact convergence rates and overall efficiency if appropriate congestion management measures are not in place. This highlights the importance of scalable, proactive congestion control strategies for large-scale robot deployments. Collectively, these studies showcase the advantages and limitations of both traditional and learning-based approaches. Building upon these insights, our research integrates biologically inspired spiral trajectories with adaptive congestion detection mechanisms, achieving a balance between structured predictability and reactive flexibility for effective swarm foraging.

III. METHODS AND DESIGN

A. Congestion Detection via Trajectory Tortuosity

The method that will be used to detect if a robot is in a “congested” state is trajectory tortuosity. Formally, tortuosity is the degree of curvature or deviation from a straight line in a trajectory [15], [16]. It is inspired by biological foraging behaviors observed in ants, bees, and other social organisms [15], [12]. In a natural environment, foragers—those hunting resources—that are experiencing local congestion often exhibit erratic and inefficient trajectories as they attempt to reach the central collection zone [12], [16], [14]. With this information, we introduce *trajectory tortuosity* as a decentralized and interpretable signal for detecting congestion in robotic swarms.

$$\text{Tortuosity} = \frac{\text{Total Path Length}}{\text{Straight-Line Distance}}$$

Trajectory Tortuosity is computed as the ratio of total distance traveled by a robot in a certain fixed window and the Euclidean distance of the first and last point of that window. Getting the result of this ratio, a straight trajectory would yield a tortuosity of 1.0 and values that are greater than 1.0 would indicate some path inefficiencies due to detours or collisions. Once the robot’s tortuosity passes a certain threshold, we can consider this robot to be in a “congested” state and it will transition to a collision/congestion avoidance behavior, such as entering a spiral pathway to alleviate local density. If a lower threshold is used like 3.0, the algorithm will be much stricter and won’t allow much deviation from the straight path. In contrast, if a 4.0 threshold is used, then restrictions are loosened and some deviation is allowed from straight path. Evaluations of optimal threshold will be tested in the experiment.

By using this biologically inspired metric to define congestion, robots are able to solely use their own motion history (x,y). This enables a fully decentralized and communication-free detection mechanism, which is crucial in large-scale swarm robotic systems operating in dynamic environments.

B. Circular Spiral Path

The circular spiral path provides a smooth, continuously curved route for robots returning to the central collection zone. A key parameter in generating this path is its smoothness, which is determined by the number of discrete steps. A higher number of steps results in a smoother spiral but also requires the robot to navigate through more waypoints, leading to increased redirection. We selected 20 steps as an effective compromise - offering a visually smooth path while minimizing computational and navigational overhead (see Fig. 2). This path is then generated by interpreting the step along the trajectory as a normalized variable, *percentage*, which varies from 0 to 1. This range is divided evenly by the step parameter, so 20 increments. The *entry* (inward) spiral can be conceptually described by the equations

$$\begin{aligned} x &= \alpha(1 - p) \cos(4\pi p + t), \\ y &= \alpha(1 - p) \sin(4\pi p + t), \end{aligned} \quad (1)$$

where p is the percentage of completion, t is a time offset allowing for dynamic adjustment, and α scales the spiral relative to the arena size. The factor 4π ensures that the path makes two full rotations as p increases from 0 to 1, thus creating a smooth inward spiral that minimizes abrupt directional changes.

The initial scope of the project included an exit spiral as well, which was just an inverse of the entry spiral. However, we found that 2 spirals, entry and exit, prove to be a great obstacle to robots that are not congested and simply trying to reach the central collection zone via their normal, straight path. For this reason, the project was conducted using only a dynamic entry path (activated locally when congested). Instead of a circular or square spiral exit path, 4 exit points were identified aligned with the cardinal directions (N, E, S, W) (see the arrows in Figs. 3 and 2). This approach also incorporates the site fidelity and pheromone trail information. For example, if a robot is using site fidelity (going back to a previous pickup location), it will take the 4 exit points and compare distances to check which one is closest to its target. It will then go to the closest exit point first and then to its initial target. Similarly, when a robot picks up a resource, its position is compared to the available entry points, and the closest one is chosen for entry. This distributed entry/exit framework helps balance the flow of robots, preventing a high concentration of traffic at a single point and thus mitigating congestion.

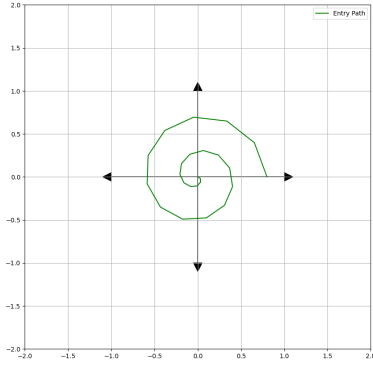


Fig. 2. Circular Spiral Path in an 8×8 grid.

C. Square Spiral Path

As shown in Fig. 3, the square spiral path is generated by a discrete step-wise process in which the robot follows a series of linear segments that change direction in a fixed sequence (e.g., right, up, left, down). At the start, the robot is positioned at the central collection zone (considered the origin, which is 0,0), and the trajectory expands outwards in

a square-like pattern. The step sizes are determined such that the total traversal remains within a predefined boundary (for instance, a 1×1 square around the central collection zone).

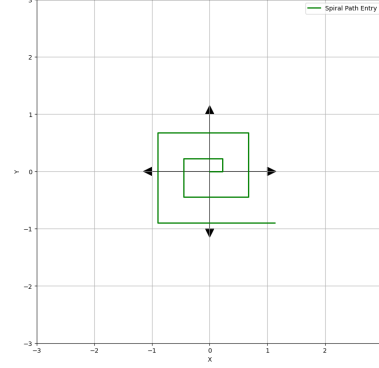


Fig. 3. Square Spiral Path in an 8×8 grid.

D. Integration of Multiple Entry/Exit Points

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E. Adaptive Congestion-Avoidance Path

In addition to these pre-planned spiral paths, an adaptive path is also constructed. In this simple algorithm, robots continuously monitor for congestion based on the *trajectory tortuosity*. When a robot is detected as congested, it will take an adaptive action and turn 90 degrees left and go 0.3 units straight. Once it finishes this action, it will attempt to reach the central collection zone and repeat the cycle (see Fig. 4).

IV. RESULTS

To evaluate the effectiveness of the proposed congestion mitigation strategies, we conducted experiments across three major categories of return path strategies: square spiral, circular spiral, and adaptive rerouting — each tested with

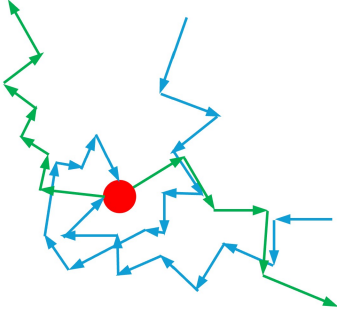


Fig. 4. Adaptive path where robot sets is target to the left, travels to this target, and continues attempt to reach central collection zone.

trajectory tortuosity thresholds of 3.0, 3.5, and 4.0. A lower threshold is stricter, labeling more movement as “congested,” while higher thresholds are more permissive. The baseline represents the default CPFA behavior without congestion mitigation. Each setting was evaluated over 50 ARGoS simulation runs in an 8×8 m arena with 32 robots. Each simulation ran for a duration of 5 minutes. Our videos are available on Youtube¹.

Table I summarizes the performance based on two metrics: the average number of resources collected and the total time spent in collisions (measured in seconds).

TABLE I

PERFORMANCE ACROSS STRATEGIES AND TORTUOSITY THRESHOLDS

Strategy	Resources Collected	Collision Time (s)
Square Spiral (T=3.0)	27.73	859.86
Square Spiral (T=3.5)	27.30	795.26
Square Spiral (T=4.0)	28.97	809.04
Circular Spiral (T=3.0)	26.91	816.18
Circular Spiral (T=3.5)	27.27	836.40
Circular Spiral (T=4.0)	26.77	838.61
Adaptive (T=3.0)	16.57	895.37
Adaptive (T=3.5)	15.63	947.51
Adaptive (T=4.0)	16.07	898.94
Baseline	12.17	1076.67

V. DISCUSSION

Our results reveal clear patterns across congestion detection strategies and thresholds. The most effective approach—square spiral ($T = 4.0$) - yielded the highest throughput (28.97 resources collected), more than doubling the baseline, as shown in Table I. The same strategy ($T = 3.5$) also achieved the lowest average collision time (795.26s), as shown in Figure 6, demonstrating that both efficiency and safety can be improved through dynamic spiral paths.

Across all strategies, a tortuosity threshold of 4.0 generally produced better throughput (see Fig. 5). This more relaxed threshold allows robots to tolerate some path irregularity before rerouting, avoiding overly aggressive responses. In

¹https://youtube.com/playlist?list=PL5F6SFAPM9QqaUAD5reIQs_ZJP4GFGVSw&si=ZZ1EAVCflqWGS-f1

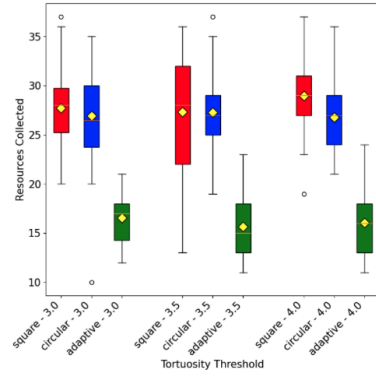


Fig. 5. Resources collected with circular, square spirals, and adaptive spiral path algorithms.

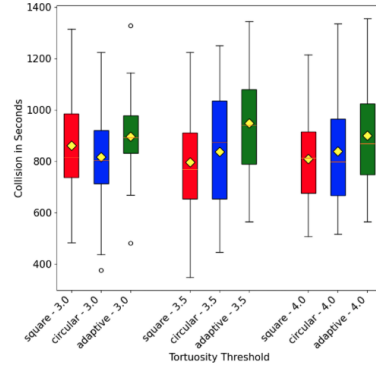


Fig. 6. Collision time with circular, square spirals, and adaptive spiral path algorithms.

contrast, stricter thresholds (like 3.0) often triggered rerouting too frequently - even in scenarios where movement was only slightly perturbed—causing unnecessary deviations and diminishing performance. This effect is especially visible in the adaptive strategy, where stricter thresholds led to more reactivity but less resource collection.

Overall, square spirals outperformed circular and adaptive strategies. We hypothesize this is due to alignment: square paths better match the arena’s grid-based layout and CPFA behavior, allowing smoother turns and quicker returns. Circular paths, while smoother in theory, introduced more curvature and directional adjustment, increasing movement time. This is supported by the slightly elevated collision times for circular paths in Figure 6, which remained consistently above 800 seconds across all thresholds. Furthermore, adaptive path rerouting, although flexible, underperformed in both metrics due to a lack of structure and coordination that is present in its spiral paths counterparts. Figure 6 shows how it exhibited some of the highest collision times, regardless of threshold value. These findings suggest that while adaptivity is important, it must be paired with organization, like spiral paths, to improve performance.

The baseline CPFA showed the worst performance: only 12.17 mean resources collected and over 1076 seconds of collision time, reinforcing the need for congestion detection. These findings highlight that structured yet decentralized

strategies - especially when tuned properly - can greatly enhance swarm performance in dense environments.

VI. FUTURE WORK

While this study does demonstrate the effectiveness of congestion-aware path planning with adaptive spiral strategies, one limitation is that these paths are static and pre-planned. In the future, we will extend our work by using Multi-Agent Reinforcement Learning (MARL) so robots can learn their adaptive behaviors that are optimized for dynamic foraging swarms, such as this environment.

The goal is for each robot to learn its policy that would minimize collisions and also reduce the amount of time taken to reach the central zone. Learning will only take place in the return phase (similar to this work), which is after a robot has picked up a resource.

The main challenge of this MARL approach is the non-stationary training. For instance, as each robot updates its policy, the environment will change for the others, and this loop will continue. This makes convergence difficult, if not impossible. To address this, we plan to adopt a Centralized Training with Decentralized Execution (CTDE) paradigm. During training, robots will share contextual information—such as positions, return duration, and congestion-related metrics—allowing for stable policy learning. During execution, each robot will act independently using only local state information, preserving scalability and robustness in the field.

We will use the Actor-Critic architecture, with the local actor policy representing an individual robot and the global critic evaluating the system's overall efficiency. The local actor state will include features such as angular deviation, time spent returning, collision count, and path efficiency relative to the optimal straight-line return. The global critic will aggregate swarm-level metrics, including the total number of robots near the nest, mean path deviation, and total collisions. These features allow the critic to evaluate the collective impact of individual behaviors, guiding the actor toward socially efficient decisions.

This learned policy may outperform the adaptive spiral by adjusting over time based on local congestion and the swarm state. In contrast to spiral and adaptive paths, MARL could discover emergent routing behaviors - such as implicit yielding, queue-forming, or decentralized avoidance - that improve throughput while maintaining collision safety. Furthermore, there is a possibility that this learning-based approach may generalize across different arena sizes, robot densities, and resource layouts, pushing toward more robust, field-ready autonomy in swarm systems.

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